

Lesson 03 - Decimal expansion representation of real numbers.

3.1 Introduction

In Lesson 02, you already learned how to represent any positive integer in any base b , where $b \geq 2$ is an integer. In this lesson you will learn, how to represent any real number as a base 10 representation. Everything you learn in this lesson can be generalized for any base b , where $b \geq 2$ is an integer. From your school days you are well familiar with this decimal expansion representation (base 10 representation). For example, you are familiar with terminating decimal expansion of rational numbers like $\frac{7}{4}$. $\frac{7}{4} = 1.75$. Also $\frac{1}{3}$ has a non-terminating decimal expansion, $\frac{1}{3} = 0.33333 \dots$. From this lesson we are studying this representations in a rigorous basis. Furthermore you will learn how to categorize real numbers by studying the nature of their decimal expansion representation. Also you will learn how to convert a terminating decimal expansion representation into the relevant rational number and how to convert recurring decimal expansion representation into the relevant rational number.



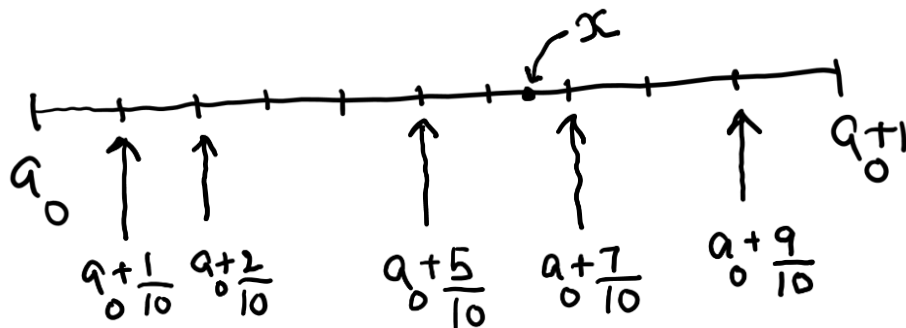
3.2 Theorem

Every positive real number x has a terminating decimal expansion $x = a_0.a_1a_2 \dots a_k$ or has a non terminating decimal expansion $x = a_0.a_1a_2 \dots a_i \dots$ where $a_0 \in \mathbb{N} \cup \{0\}$, $a_i \in \{0, 1, 2, \dots, 9\}$ for each $i \in \mathbb{N}$.

Proof: -

Observe that in the real line x lies between two consecutive integers a_0 & a_0+1 . Where a_0 is the lowest of these so that $a_0 \leq x < a_0+1$.

Now divide the line segment $[a_0, a_0+1]$ in to 10 equal parts. Then x lies in one of these sections.



Hence we can find a_1 where, $a_1 \in \{0, 1, \dots, 9\}$

such that $a_0 + \frac{a_1}{10} \leq x < a_0 + \frac{a_1}{10} + \frac{1}{10}$.



Similarly next divide the line segment $\left[a_0 + \frac{a_1}{10}, a_0 + \frac{a_1+1}{10} \right]$ into 10 equal parts

and see where x lies. Hence find a_2 such that $a_0 + \frac{a_1}{10} + \frac{a_2}{10^2} \leq x < a_0 + \frac{a_1}{10} + \frac{a_2+1}{10^2}$.

This process can be stopped, if there exist some $i \in \mathbb{N} \cup \{0\}$ such that $x = a_0 + \frac{a_1}{10} + \dots + \frac{a_i}{10^i}$.

Otherwise process continues indefinitely.

In the case that process terminates at i^{th} step, $x = a_0 + \frac{a_1}{10} + \dots + \frac{a_i}{10^i}$ — (1)

and if process continues indefinitely we have $x = a_0 + \frac{a_1}{10} + \dots + \frac{a_i}{10^i} + \dots = \sum_{i=0}^{\infty} \frac{a_i}{10^i}$ — (2)

In shorter notation we denoted these two as,

$$x = a_0.a_1a_2\dots a_i$$

$$x = a_0.a_1a_2\dots a_i\dots$$



? Question

For which real numbers, the decimal expansion representation has the form ①, and for which real numbers, the decimal expansion has the form ②.

3.3 Recurring decimals

In the case of process continuous indefinitely, this is if the real number has infinite decimal representation, there are two possibilities. either the sequence of terms $\langle a_i \rangle$ is periodic a_k or decimal expansion is not periodic for ever. In the first case we say decimal expansion is recurring. In the second case we say decimal expansion is non-recurring.

Two examples for the first case is,

$$0.65656565 \dots \quad \text{————— (1)}$$

$$0.123178910232232232 \dots \quad \text{————— (2)}$$

Observe that in (1) 65 is recurring from the beginning, but in (2) 232 is recurring after sum decimals. whenever the decimal expansion is recurring we can use the following notation

$$0.65656565 \dots = 0.\overline{65}$$

$$0.123178910232232232 \dots = 0.123178910\overline{232}$$

Exercise :- Using this notation, write down the decimal expansion representation of following rational numbers.

(i) $\frac{1}{11}$

(ii) $\frac{22}{7}$

(iii) $\frac{5}{34}$

(iv) $\frac{43}{13}$



An example for decimal expansion is not recurring for ever is,

0.101001000100001000001... $\leftarrow$ (This pattern continues)

3.4 Theorem

The decimal expansion of non negative real number x will terminates if and only if x is a rational number, whose denominator is of the form $2^n 5^m$, where m and n are non negative integers.

Proof :-

Suppose decimal expansion of x is terminating.

Therefore $x = \sum_{i=0}^n \frac{a_i}{10^i}$ for some $n \in \mathbb{N}$.

$$\begin{aligned} \text{This is } x &= a_0 + \frac{a_1}{10} + \dots + \frac{a_n}{10^n} \\ &= \frac{a_0 10^n + a_1 10^{n-1} + \dots + a_n}{10^n} \end{aligned}$$

Note that numerator is an integer and denominator is of the form $2^n 5^n$.



Conversely, suppose the denominator is of the form $2^n 5^m$ for some non negative integers m and n .

Hence $x = \frac{p}{2^n 5^m}$ where $p \in \mathbb{N}$.

$$\therefore x = \frac{2^m 5^n p}{2^n 5^m \cdot 2^m 5^n} = \frac{2^m 5^n p}{10^{m+n}}$$

Now observe that $2^m 5^n p$ is a positive integer

$\therefore$ it can be represent in the form

$$a_0 + a_1 10 + a_2 10^2 + \dots + a_k 10^k.$$

$$\text{Hence } x = \frac{2^m 5^n p}{10^{m+n}} = \frac{a_0 + a_1 10 + a_2 10^2 + \dots + a_k 10^k}{10^{m+n}}$$

$$= \frac{a_k 10^k}{10^{m+n}} + \frac{a_{k-1} 10^{k-1}}{10^{m+n}} + \dots + \frac{a_1 10}{10^{m+n}} + \frac{a_0}{10^{m+n}}$$

$\longrightarrow \textcircled{\alpha}$



Now observe that, $\frac{a_k 10^{k-2^0}}{10^{m+n}}$ is an integer

if $k-2^0 \geq m+n$. Hence the expansion of (4)

can be represented as,

$$4 + \frac{a_j}{10} + \frac{a_{j-1}}{10^2} + \dots + \frac{a_1}{10^{m+n-1}} + \frac{a_0}{10^{m+n}}$$

where 4 is an integer.

Hence x has a terminating decimal expansion. Hence the theorem.

Exercise:-

(a) Find the decimal expression of following real numbers

(i) $\frac{7}{5^2}$ (ii) $\frac{9}{256}$ (iii) $\frac{11}{160}$ (iv) $\frac{259}{2^{10} \cdot 5^{13}}$

(b) Find the rational number relevant to following decimal expansions

(i) 7.3332 (ii) 0.1414 (iii) 0.0019531



3.5 Theorem

If a rational number does not have a denominator of the form $2^m 5^n$, where m, n are non negative integers, then the decimal expansion of that rational number is periodic from some decimal.

This is it has decimal expansion of the form

$$a_0 : a_1 a_2 \dots a_m \overline{a_{m+1} a_{m+2} \dots a_{m+p}} \text{ for some } m \in \mathbb{N} \cup \{0\} \text{ and } p \in \mathbb{N}.$$

Proof - Omit.

(You need more advance theorems to Prove this)

Now let's prove the converse of this theorem.

3.6 Theorem

If the decimal expansion of a real number has a form

$$a_0 : a_1 a_2 \dots a_m \overline{a_{m+1} a_{m+2} \dots a_{m+p}} \text{ for some } m \in \mathbb{N} \cup \{0\} \text{ and } p \in \mathbb{N}$$

then it is a rational number.



Proof:-

Observe that

$$a_0 a_1 \dots a_m \overline{a_{m+1} a_{m+2} \dots a_{m+p}} = a_0 a_1 a_2 \dots a_m + \left(\overline{a_{m+1} a_{m+2} \dots a_{m+p}} \right) \frac{1}{10^m}$$

$$\text{But } a_0 a_1 a_2 \dots a_m = \frac{a_0 a_1 a_2 \dots a_m}{10^m}$$

Hence $a_0 a_1 a_2 \dots a_m$ is a rational number.

Therefore it is enough to show that,

$0.\overline{a_{m+1} a_{m+2} \dots a_{m+p}}$ is a rational number (why?)

Observe that,

$$0.\overline{a_{m+1} a_{m+2} \dots a_{m+p}} = \frac{a_{m+1}}{10} + \frac{a_{m+2}}{10^2} + \dots + \frac{a_{m+p}}{10^p} + \frac{a_{m+1}}{10^{p+1}} + \frac{a_{m+2}}{10^{p+2}} + \dots$$

$$= \frac{1}{10} \left(a_{m+1} + \frac{a_{m+2}}{10} + \dots + \frac{a_{m+p}}{10^{p-1}} \right) + \frac{1}{10^{p+1}} \left(a_{m+1} + \frac{a_{m+2}}{10} + \dots + \frac{a_{m+p}}{10^{p-1}} \right) + \dots$$

$$= \left(a_{m+1} + \frac{a_{m+2}}{10} + \dots + \frac{a_{m+p}}{10^{p-1}} \right) \sum_{i=0}^{\infty} \frac{1}{10^{p+i}}$$



$$= \left(a_{m+1} + \frac{a_{m+2}}{10} + \dots + \frac{a_{m+p}}{10^{p-1}} \right) \frac{\frac{1}{10}}{1 - \frac{1}{10^p}}$$

$$= \frac{10^{p-1}}{(10^p - 1)} \left(a_{m+1} + \frac{a_{m+2}}{10} + \dots + \frac{a_{m+p}}{10^{p-1}} \right)$$

Which is a rational number. Hence the theorem.

Exercise:-

Find the relevant rational numbers for the following decimal expansions.

(i) $0.\overline{7}$ (ii) $0.\overline{48}$ (iii) $0.553\overline{17}$

(iv) $0.\overline{0588235294117647}$

(v) $0.\overline{123456789}$ (vi) $7.\overline{9}$



3.6 A shorter method to find relevant rational number of a recurring decimal expansion. (Periodic decimal)

Let's demonstrate this from some concrete examples.
Let's find the relevant rational numbers for the $0.\overline{01}$ and $0.2\overline{345}$.

First find the relevant rational number of $0.\overline{01}$, by using the technique used in theorem 3.5.

$$\begin{aligned} 0.\overline{01} &= (\overline{01}) = \left(\frac{0}{10} + \frac{1}{10^2} + \frac{0}{10^3} + \frac{1}{10^4} + \frac{0}{10^5} + \dots \right) \\ &= \left(\frac{1}{10^2} + \frac{1}{10^4} + \frac{1}{10^6} + \dots \right) = \frac{1}{10^2} \left(1 + \frac{1}{10^2} + \frac{1}{10^4} + \dots \right) \\ &= \frac{1}{10^2} \left(\frac{1}{1 - \frac{1}{10^2}} \right) = \frac{100}{10^2 \cdot 99} = \frac{1}{99} \end{aligned}$$

Also without referring to geometric series we can use following method.

$$x = 0.\overline{01} \quad \text{--- (1)}$$

$$\therefore 100x = 1.\overline{01} \quad \text{--- (2)}$$

$$\textcircled{2} - \textcircled{1} \quad 99x = 1 \quad \therefore x = 1/99$$



Why this trick works? You must not use any trick without explanation. Let's find the rational behind this trick

in (1) $\overline{01}$ represent the summation of convergent infinite series. In (2) $\overline{01}$ again represent the summation of same convergent series. By subtracting $\times 04$ cancelled the summation. So this trick allowed to you to find the value of x without finding the summation of the convergent series.

Let's use this trick to find the rational number relevant to $0.2\overline{345}$

$$\text{Let } x = 0.2\overline{345}$$

$$\therefore 10x = 2.\overline{345} \quad \text{--- (1)}$$

$$10^4 x = 2345.\overline{345} \quad \text{--- (2)}$$

$$\text{By (2) - (1) } 9990x = 2343$$

$$\therefore x = \frac{2343}{9990} = \frac{781}{3330}$$



? Question

Are there any two different decimal representations which represent same rational number?

Note: Observe that, through out, this lesson we developed the theory only for the decimal representations. But notice that everything can be generalized to representations for any basis.

Review questions.

(1) In the binary representation some rational numbers can be represent as

(i) $0.101101101101\dots$ (ii) $0.11111111\dots$

Find the relevant rational numbers.
(write your answer in decimal forms)

(2) Note that $\frac{1}{7}$ has a periodic decimal expansion with minimal period 6. Are there any odd prime p such that decimal expansion of $\frac{1}{p}$ has period $p-1$?

(3) Is there any positive integer m such that $\frac{1}{m}$ has a periodic decimal expansion with minimal period greater than or equal to m ? Justify your answer?

(4) Following chart classify the decimal expansions of real numbers.

